

task:-3

**Deploye your flask backend and express
frontend docker container using aws
ecr,ecs and vpc service**

EC2

Dashboard

AWS Global View

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Instances

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Instances (1/4)

Info

Find Instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
☐	flask-backend	i-06c8c31d2079d401f	Stopped	t2.micro	–	us-east-1c	–	–	–	–
☐	express-frontend	i-02eb409fb6f542845	Stopped	t2.micro	–	us-east-1c	–	–	–	–
☐	flask-express-s...	i-09001c37363bd2345	Stopped	t2.micro	–	us-east-1c	–	–	–	–
☒	docker-server	i-01fbf6a30eb8973c4	Running	t2.micro	Initializing	us-east-1c	ec2-13-222-0-111.comp...	13.222.0.111	–	–

i-01fbf6a30eb8973c4 (docker-server)

Details

Status and alarms

Monitoring

Security

Networking

Storage

Tags

Instance summary

Instance ID

i-01fbf6a30eb8973c4

IPv6 address

–

Hostname type

IP name: ip-172-31-20-40.ec2.internal

Answer private resource DNS name

IPv4 (A)

Auto-assigned IP address

13.222.0.111 [Public IP]

IAM role

Public IPv4 address

13.222.0.111 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-172-31-20-40.ec2.internal

Instance type

t2.micro

VPC ID

vpc-0e518e59086e169b7

Subnet ID

Private IPv4 addresses

172.31.20.40

Public DNS

ec2-13-222-0-111.compute-1.amazonaws.com | open address

Elastic IP addresses

–

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

us-east-1.console.aws.amazon.com / EC2 Instance Connect

aws

Search

[Alt+S]

United States (N. Virginia)

Shakti Kumar Behera (893675906830)

Shakti Kumar Behera

----- 226.5/226.5 kB 197.3 MB/s eta 0:00:00
Downloading packaging-26.2-py3-none-any.whl (100 kB)
----- 100.2/100.2 kB 170.8 MB/s eta 0:00:00
Downloading markupsafe-3.0.3-cp311-cp311-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (22 kB)
Installing collected packages: packaging, MarkupSafe, itsdangerous, click, blinker, Werkzeug, Jinja2, gunicorn, Flask
Successfully installed Flask-3.0.3 Jinja2-3.1.6 MarkupSafe-3.0.3 Werkzeug-3.1.8 blinker-1.9.0 click-8.4.2 gunicorn-22.0.0 itsdangerous-2.2.0 packaging-26.2
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv

[notice] A new release of pip is available: 24.0 -> 26.1.2
[notice] To update, run: pip install --upgrade pip
---> Removed intermediate container af38fa68bf65
---> c84c9a969923
Step 5/7 : COPY . .
---> 77fd4aef5813
Step 6/7 : EXPOSE 5000
---> Running in 72511839aaf4
---> Removed intermediate container 72511839aaf4
---> bd00820406f6
Step 7/7 : CMD ["python", "app.py"]
---> Running in 0d47fa1bfe85
---> Removed intermediate container 0d47fa1bfe85
---> c1473e458457
Successfully built c1473e458457
Successfully tagged flask-app:latest
ubuntu@ip-172-31-20-40:~/Kubernetes-Project/frontend\$ docker images

Info →

U In Use

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
express-app:latest	b513c6e3ca49	1.59GB	401MB	
flask-app:latest	c1473e458457	213MB	51.6MB	
node:20	8f693eaa7e0a	1.59GB	415MB	
python:3.11-slim	b27df5841f33	188MB	47.7MB	

ubuntu@ip-172-31-20-40:~/Kubernetes-Project/frontend\$ docker run -d --name express-container -p 3000:3000 express-app
b2bcab09b660a4bb5ad6f81cef864b52d381cabb3436e555ae888931f3a7dd3c
ubuntu@ip-172-31-20-40:~/Kubernetes-Project/frontend\$ docker ps

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
b2bcab09b660	express-app	"docker-entrypoint.s..."	About a minute ago	Up About a minute	0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp	express-container

ubuntu@ip-172-31-20-40:~/Kubernetes-Project/frontend\$ docker run -d --name flask-container -p 5000:5000 flask-app
83ed51367da9097793efc689af656ea908dd537be90b384484a3642d39608202
ubuntu@ip-172-31-20-40:~/Kubernetes-Project/frontend\$ docker ps

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
83ed51367da9	flask-app	"python app.py"	8 seconds ago	Up 7 seconds	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp	flask-container
b2bcab09b660	express-app	"docker-entrypoint.s..."	2 minutes ago	Up 2 minutes	0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp	express-container

ubuntu@ip-172-31-20-40:~/Kubernetes-Project/frontend\$ cd ..
ubuntu@ip-172-31-20-40:~/Kubernetes-Project\$

i-01fbf6a30eb8973c4 (docker-server)

us-east-1.console.aws.amazon.com / Elastic Container Registry

aws

Search

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United States (N. Virginia)

Shakti Kumar Behera (893675906830)

Shakti Kumar Behera

Amazon ECR

Private registry

Repositories

Amazon Elastic Container Registry

Private registry

Repositories

Features & Settings

Public registry

Repositories

Settings

ECR public gallery

Amazon ECS

Amazon EKS

Getting started

Documentation

Private repositories (2)

View push commands

Delete

Actions

Create repository

Search by repository substring

Repository name	URI	Created at	Tag immutability	Encryption type
express-app	893675906830.dkr.ecr.us-east-1.amazonaws.com/express-app	July 03, 2026, 21:36:48 (UTC+05.5)	Mutable	AES-256
flask-app	893675906830.dkr.ecr.us-east-1.amazonaws.com/flask-app	July 03, 2026, 21:37:25 (UTC+05.5)	Mutable	AES-256


```

Receiving objects: 100% (17/17), 470.11 KiB | 17.41 MiB/s, done.
Resolving deltas: 100% (1/1), done.
ubuntu@ip-172-31-20-40:~$ cd Kubernetes-Project
ubuntu@ip-172-31-20-40:~/Kubernetes-Project$ ls
Kubernetes-Assignments.pdf  backend  frontend  k8s
ubuntu@ip-172-31-20-40:~/Kubernetes-Project$ cd backend
ubuntu@ip-172-31-20-40:~/Kubernetes-Project/backend$ docker build -t express-app
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

```

Docker: 'docker build' requires 1 argument

```
Usage:  docker build [OPTIONS] PATH | URL | -
```

Run 'docker build --help' for more information

```
ubuntu@ip-172-31-20-40:~/Kubernetes-Project/backend$ docker images
```

IMAGE	ID	DISK_USAGE	CONTENT_SIZE	EXTRA
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```
ubuntu@ip-172-31-20-40:~/Kubernetes-Project/backend$ docker build -t express-app .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/
```

```

sending build context to Docker daemon 4.096kB

```

```
Step 1/7 : FROM node:20
```

```
0: Pulling from library/node
```

```
b22ec2d9b1a: Pulling fs layer
```

```
b53381ee51f: Pulling fs layer
```

```
5203b2bfeff: Pulling fs layer
```

```
fcda2b4d799: Pulling fs layer
```

```
933ecab0f11: Pulling fs layer
```

```
83419f44309: Pulling fs layer
```

```
5e6b8204a89: Pulling fs layer
```

```
f2ab6e4eb46: Pulling fs layer
```

```
e22c7cc721e: Download complete
```

```
b22ec2d9b1a: Download complete
```

```
E2ab6e4eb46: Download complete
```

```
83419f44309: Download complete
```

```
4b081db89c0: Download complete
```

```
5203b2bfeff: Download complet
```

```
b53381ee51f: Download complete
```

```
fcda2b4d799: Download complete
```

```
5e6b8204a89: Download complete
```

i-01fbf6a30eb8973c4 (docker-server)

PublicIPs: 13.222.0.111 PrivateIPs: 172.31.20.40

THANK YOU!

Feel free to approach us if you have any questions.